# Physics Project Script – Automatic Doorbell System

Presented by:
Members:
Class:
Exhibition:

## 🎬 Introduction

Good morning respected teachers, mentors, and fellow students.

We are The Creative Crew, a group of Class 10 students — Nitya Singh, Yashita Singh, and Rajveer Singh.

Let us take you back to the time of the COVID-19 pandemic. It taught us many things — one of the most important being the value of contactless interaction.

Even now, as new cases rise again in some places, the importance of hygiene and safety remains strong.

That's why, inspired by this, we have built a Contactless Automatic Doorbell System using IR Sensor and Arduino — a project that combines safety with science, based on simple Class 10 physics concepts.

## 🛠️ Working Explained

Our system works in 3 simple steps:

1. IR Sensor detects when someone comes close to the door using invisible infrared light.
2. Arduino UNO receives this signal and controls when the doorbell should ring.
3. ISD1820 Voice Module plays a recorded sound or message to inform people inside.

This project may look modern, but it is based on basic physics:

- Reflection of Light: IR sensor detects reflected infrared rays.
- Electricity and Circuits: All components are powered and connected electrically.
- Sound Production: Voice module turns electric signals into sound.
- Energy Conversion: From electrical energy to sound energy.

- Logic and Delay: Used to control when and how often the bell rings.

## 🧠 Real-Life Use & Conclusion

This project can be used:
- In homes for smart entry systems
- In hospitals to avoid contact-based bells
- In offices, schools, and during health emergencies

It's a perfect mix of science, safety, and smart thinking.

Thank you for encouraging innovation.
We hope you liked our idea. We're ready for your questions!

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We've also programmed it smartly:
- It won't ring again and again if the same person is still standing.
- If no one opens the door within 5 seconds, it rings again as a reminder.

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